Rakesh Lakshmanan

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rakeshlakshmanan.github.io/portfolio

Summary

Backend engineer with 2+ years building Java platform infrastructure at Zoho (HA/DR, database backends, and migration tools), plus Spring Boot and FastAPI microservices on Kubernetes. MSc Computer Science graduate from Trinity College Dublin (1:1, Distinction, Aug 2026). **Work Authorisation: Stamp 1G (Irish Third Level Graduate Visa)**.

Experience

Member Technical Staff (Software Engineer) Jun 2023 - Aug 2025
Zoho Corporation *Chennai, India*

- Won Zoho's Mickey Emerging Programmer award (Q4 FY23) for building reusable components in Mickey, a Java data-access framework that replaces hand-written SQL with one query API across PostgreSQL and MSSQL and handles schema migrations
- Delivered 99.95% uptime with a four-node HA/DR topology, config-driven scheduling, and Robocopy replication
- Removed ∼60% of manual work from upgrades and failovers and cut recovery time by ∼35% by automating production server upgrades with backup, health checks, and rollback
- Reduced typical analytic query time from ∼2.1 s to ∼0.15 s and storage by ∼10x for analytics-heavy products by adding ClickHouse as a columnar backend behind the framework's Java query API
- Shortened release time by 30% by developing test-run and reporting features for Zoho's QA platform in Java/Spring Boot, running suites through Jenkins pipelines in Docker and indexing test, JaCoCo, and SonarQube reports in Elasticsearch
- Moved ∼80% of query traffic off the primary database by adding read-only server support to the framework's data layer, routing read queries to replicas
- Cut the time to find migration differences by 90% by writing a Java tool that compares the schemas and data of two databases over JDBC and reports every difference
- Fixed 200 long-standing defects with QA through to root cause and mentored two interns on HA and fault tolerance

Project Trainee Jan 2023 - May 2023
Zoho Corporation *Chennai, India*

- Cut manual testing effort by 60% and expanded regression coverage by 40% by architecting a test automation framework for Java Swing desktop applications
- Lowered file-transfer error rates from 22% to 7% by building a Java NIO utility that keeps files in sync between primary and backup servers
- Replaced per-application HDFS client code with one shared Java NIO filesystem provider, letting apps read and write HDFS through the standard `java.nio.file` API

Summer Intern Apr 2022 - Sep 2022
Zoho Corporation *Chennai, India*

- Closed 500+ production and support tickets and shipped 100+ bug fixes in six months; built a SQLite-backed FIFO queue in Java for task and message processing

Projects

MSc Dissertation: Modern Delay-Tolerant Email
Docker, Postfix, Thunderbird, ION-DTN, BPMail (Bundle Protocol), Linux

- Studied email and anti-spam tooling under cislunar delays, supervised by Stephen Farrell (co-author of RFC 9174)
- Set up a Docker testbed to test mail delivery and anti-spam checks over simulated cislunar DTN links

DevVault Live | GitHub
Java 17, Spring Boot, Spring Security, PostgreSQL, Gemini 2.5 Flash, React, Docker, Kubernetes (GKE), GitHub Actions

- Deployed a bookmark manager on GKE that tags and summarises saved pages, with JWT and Google OAuth2 SSO
- Built an event-driven AI pipeline with Spring transactional events and a dedicated thread pool, so bookmarks are saved instantly and enriched by Gemini in the background

SmartEnoughCity GitHub Oct 2025 - Mar 2026
Java 17, Spring Boot, FastAPI, scikit-learn, React, PostgreSQL/PostGIS, GKE, Helm, Terraform, Istio, GitHub Actions

- Developed a 4-service platform ingesting 15+ Dublin transport APIs into PostGIS for scikit-learn forecasts
- Provisioned GKE with Terraform, Helm, and Istio; GitHub Actions runs ∼600 tests with a 75% coverage floor

Distributed Traffic Booking System GitHub Sep 2025 - Dec 2025
Python 3.12, FastAPI, SQLAlchemy (async), PostgreSQL, Redis, RabbitMQ, React, TypeScript, Docker, AWS (EC2, RDS)

- Architected a 6-service booking platform over Ireland, UK, and France partitions with Saga orchestration
- Prevented double-booking with Redis locks per road segment and time slot, plus a 120s saga reconciler

Education

Trinity College Dublin Dublin, Ireland
MSc in Computer Science, Future Networked Systems | 1:1, Distinction *Sep 2025 - Aug 2026*

St. Joseph's College of Engineering Chennai, India
Bachelor's in Computer Science | 1:1 (92.3%) | ELGI Scholar *Jun 2019 - Apr 2023*

Technical Skills

Languages & Tools: Java (JDK 17), Python, TypeScript, JavaScript, SQL, C, Git, Gradle, Maven, Linux, Claude Code, Copilot
Frameworks & APIs: Spring Boot, Spring Security, FastAPI, Node.js, React, REST APIs, microservices, Gemini API
Distributed Systems: HA/DR, fault tolerance, replication, Sagas, distributed locks, event-driven architecture, RabbitMQ
Cloud & DevOps: AWS (EC2, RDS), GKE, Docker, Kubernetes, Helm, Terraform, Istio, GitHub Actions, Prometheus, Grafana
Data & Networking: PostgreSQL (PostGIS), MySQL, MSSQL, ClickHouse, MongoDB, Redis, TCP/IP, DNS, HTTP, DTN